

WILP, BITS Pilani

MACHINE LEARNING · COMPANION READER

Instance-Based Learning & k-NN

Lazy vs Eager Learning, Distance Metrics, k-NN Regression, Distance Weighting, and Dimensionality

Session 9 · Read alongside the lecture slides · Every slide example worked out in full

Prepared for the Work Integrated Learning Programmes (WILP), BITS Pilani.

1 Lazy vs Eager Learning

Lazy learners store training data and defer computation until a query arrives.

The intuition

Instance-based learning classifies a new object by looking at its most similar stored neighbors.

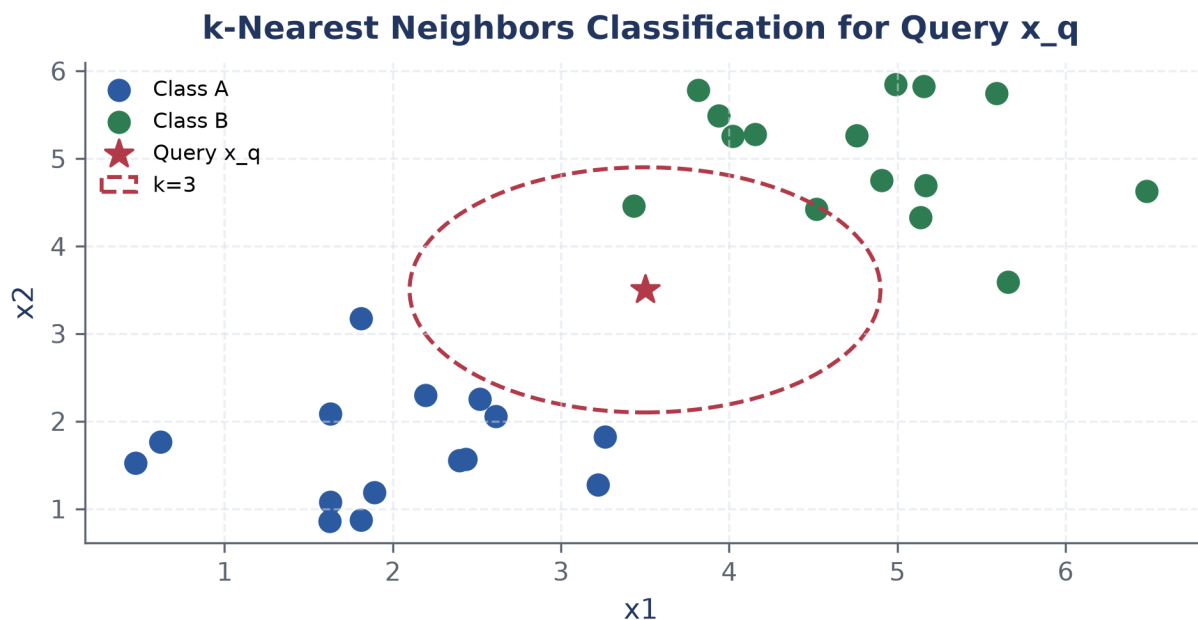


Figure 1: k-NN classification for query x_q .

2 Distance Metrics for Feature Spaces

Distance measures define the geometrical similarity between data points.

Given two d -dimensional feature vectors $\mathbf{x}$ and $\mathbf{y}$, distance metrics formalize similarity.

$$L_p(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^d |x_i - y_i|^p \right)^{1/p}. \quad (1)$$

Unit Distance Contours for Minkowski Distance Norms

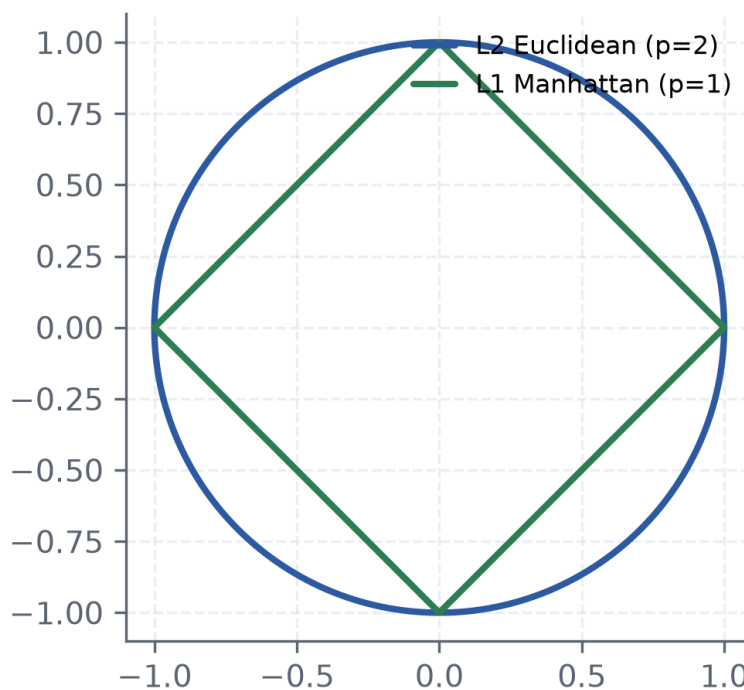


Figure 2: Unit distance contours for Manhattan (L_1), Euclidean (L_2), and Chebyshev (L_∞) metrics.

3 k-NN Classification & Regression

k-NN predicts discrete labels via majority voting and continuous targets via local averaging.

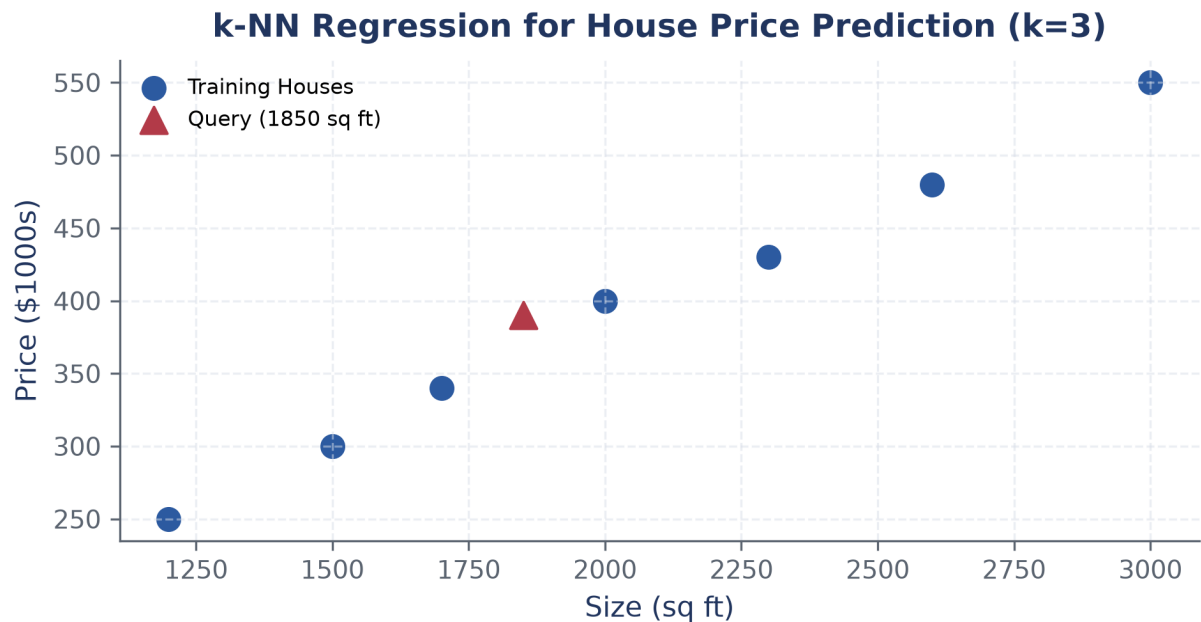


Figure 3: k-NN regression for house price prediction ($k = 3$).

 Worked example: k-NN Regression for House Price Prediction

Predict house price for size $x_q = 1850$ sq ft using $k = 3$ nearest neighbors:

Step 1. $k = 3$ nearest sizes are 1700 (\$340k), 2000 (\$400k), and 1500 (\$300k).

Step 2. Compute average price: $\hat{y} = (340 + 400 + 300)/3 = \$346.67k$.

Glossary & Symbols

Key Terms

Lazy Learning Storing training instances without explicit model training until prediction time.

k-Nearest Neighbors

A non-parametric method classifying/regressing based on k closest training points.